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MILITARY SPECIFICATION

TEST KIT, RESERVE ALKALINITY, Antifreeze

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a test kit suitable for determining the reserve alkalinity and corrosion inhibition of the coolant in liquid cooling systems.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

MILITARY

MIL-P-116	- Preservation-packaging, Methods of.
MIL-L-10547	- Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document.

FSC 6630

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout and shall be as specified herein.

3.2 Design and construction. The test kit shall consist of a glass screw cap bottle containing 25 test strips and a desiccant packet. Each test kit shall be supplied with a length of rubber tubing, a color chart and instructions.

3.2.1 Bottle and cap closure.

* 3.2.1.1 Bottle. The bottle shall be fabricated of amber glass and shall be cylindrical in shape. Bottle shall have a wide mouth opening and shall have a continuous thread finish with at least one complete turn of full thread projection. The bottom shall be so constructed that the bottle shall be stable when placed on a smooth flat surface. The bottle shall have the following minimum dimensions (inches):

Overall height----------3-5/8
Inside diameter at mouth opening----------7/8

3.2.1.2 Closure. Shall be a metal screw cap coated to resist corrosion. The screw cap shall be fitted with a liner facing impervious to water. The liner shall be secured to the inside of the cap in such a manner that it does not fall out of the cap. The liner shall be of sufficient thickness and resilience to form an effective seal with the top surface of the bottle. Closure shall be threaded with continuous type thread and shall mate with the finish of the bottle to provide a good seal with the edges of the mouth opening. Vertical ribs shall be provided on the outer walls of the screw cap to facilitate removal and tightening of the closure.

3.2.2 Test strip. The test strip shall be a plastic strip with an impregnated reagent pad at the distal end. The plastic material shall be clear, semi-rigid polystyrene measuring 3-1/4 ± 1/8 inches long by 13/64 ± 1/64 inch wide. A suitable opaque material shall be securely attached to the strip from the distal end to at least the middle of the strip. A 13/64 ± 1/64 by 13/64 ± 1/64 inch impregnated reagent pad shall be securely adhered to one end of the plastic strip.

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One side of the pad shall not be obstructed by the opaque material. The reagent pad shall consist of white filter paper, Whatman 3MM paper or equivalent, and shall be 0.33 ± 0.033 millimeter (mm) in thickness. The impregnated pad shall be capable of changing its color in accordance with 3.3.

3.2.3 Rubber tubing. A length of rubber tubing shall be furnished for use as necessary to serve as an extension and to facilitate in dipping the test strip into the radiator coolant. The rubber tubing shall be a minimum of 7-5/8 inches long with an inside diameter of 3/16 ± 1/32 inch and a minimum wall thickness of 3/64 inch.

* 3.2.4 Color chart. A color chart conforming to 3.4 shall be supplied with each test kit.

3.3 Performance. The impregnated reagent pad on the test strip shall react to various concentrations of alkalinity in cooling solutions by changing to the following colors:

a. Yellowish-green to indicate that solution has a "poor" condition for corrosion protection.
b. Darker green to indicate that solution has a "borderline" condition for corrosion protection.
c. Blue to indicate that solution has a "good" condition for corrosion protection.

When tested as specified, the color reaction on each reagent pad shall match its corresponding color standard (see 4.4.1).

3.4 Marking on bottle. Each bottle shall have front and back labels of white litho paper cast-coated on one side and shall include the following information:

Front label.

a. Item identification designated as "TEST KIT, RESERVE ALKALINITY, Anti-freeze".
b. The number of tests designated as "25 tests".
c. A statement that the contents of the kit include 25 test strips and desiccant.
d. The lot or control number.
e. Date of manufacture.
f. The name and address of the manufacturer.

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* Back label.

a. Directions to include the following:

(1) Dip test area of strip into radiator in contact with antifreeze and remove immediately.
(2) Shake excess fluid from test area.
(3) Compare color to color chart after 15 seconds.

b. Color chart for interpretation of the color reaction on the test strip consisting of 3 colors - blue, green and yellowish green.

Blue shall be designated as "good" and "Satisfactory". Green shall be designated as "Borderline" and "Satisfactory". Yellowish green shall be designated as "Poor" and "Unsatisfactory".

c. The following statements:

(1) Make sure freeze protection is not lower than -55° Fahrenheit (°F.) (see instructions).
(2) Use at temperature above 50°F. or when coolant is hot.
(3) Caution: To obtain correct color comparison, read exposed portion of the impregnated reagent pad.

3.5 Instructions. Instructions for use of the test kit shall be furnished in sheet or pamphlet form with each unit. Instructions shall include procedure for use of the test kit by way of description or simple illustrations or both. Purpose of the test kit shall be clearly presented, including information as to the satisfactory and unsatisfactory corrosion protection of the coolant in cooling systems. Information shall also be provided as to "the importance of corrosion testing" and "how the test strip reagent works" in process of dipping into coolant. The following precautions or similar statements shall also be indicated:

a. Do not use when the freeze protection of the antifreeze solution is lower than -55°F. Corrosion inhibitors are less effective in coolants having very high antifreeze concentrations. This kit is not designed to estimate corrosion protection in these solutions.
b. Test at temperature above 50°F.
c. Test indicates current corrosion protection but not how long it will last.
d. Protect reagent strips from moisture, sunlight and excessive heat. Store at controlled room temperature (59° - 86°F.).

3.6 Lot size. For purposes of this specification, a lot, batch or control shall be the single, uniform quantity of test strips produced from one formulation, subjected to the same manufacturing operation and contained in the bottle.

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3.7 Workmanship. Workmanship shall be first class throughout. Test kit shall be free from defects which detract from their appearance or impair their serviceability. The test strip shall be free of tears, cuts and folds. The bottle shall be free of cracks, chips and breaks. The bottle shall be clean and dry, and shall be free of dirt and foreign matter. The rubber tubing shall be free of tackiness and brittleness.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three (3) years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The sample unit shall be one test kit. One strip shall be selected at random from one bottle.

TABLE I. For examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. Sample unit shall be one test kit. One strip shall be selected at random from one bottle.

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TABLE II. For tests.

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level	AQL
End item	Performance	3.3	4.4.1	S-4*	**

* When necessary, additional test strips shall be selected so that an equal number of tests can be conducted with each standard solution.
** Acceptance number shall be zero.

4.3 Examination. The test kit and components shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with table III.

TABLE III. Classification of defects.

Category	Defects*
Major	
101	Component missing.
102	Marking incomplete, illegible or missing.
103	Label not completely affixed to bottle.
104	Bottle not free of cracks, chips and breaks.
105	Bottle not stable when placed on a smooth, flat surface.
106	Bottle not of amber glass.
107	Threading on bottle and closure not continuous.
108	Reagent pad not completely adhered to plastic strip.
109	Plastic strip not free of tears and cuts.
110	Plastic strip not free of folds.
111	Instructions incomplete, illegible or missing.
112	Bottle not tightly sealed.
Minor	
201	Minimum number of test strips in unit bottle not "25".
202	Bottle not clean, dry or free of dirt or foreign matter.
203	Ribs on outside of screw cap not present.
204	Rubber tubing not free of tackiness and brittleness.
205	Opaque material on plastic strip missing or incomplete.

* Examination shall not be restricted to the defects listed above.

4.3.2 Dimensional examination. The components of the test kit shall be examined for defects in dimensions. Any dimension not within specified limits shall be considered as a defect.

4.3.3 Packaging inspection. The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

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4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Performance (color reaction of reagent pad).

4.4.1.1 Preparation of standard test solutions. Three types of solutions, representing different concentrations of alkalinity in cooling systems (see 3.3), shall be prepared in accordance with the following:

a. "POOR" type solution. Weigh accurately 425 mg. of Na_2HPO_4 and dissolve in 20 milliliters (ml) of distilled water in a 100 ml. volumetric flask. Add 40 ml. of ethylene glycol. Bring to volume with distilled water.

b. "BORDERLINE" type solution. Weigh accurately 900 mg. of Na_2HPO_4 and dissolve in 20 ml. of distilled water in a 100 ml. volumetric flask. Add 40 ml. of ethylene glycol. Bring to volume with distilled water.

c. "GOOD" type solution. Weigh accurately 1500 mg. of Na_2HPO_4 and dissolve in 20 ml. of distilled water in a 100 ml. volumetric flask. Add 40 ml. of ethylene glycol. Bring to volume with distilled water.

NOTE:
1. Solutions shall be used the same day they are prepared.
2. Chemical reagents shall be of reagent grade.

After preparing the solutions, the test strips shall be tested in accordance with 4.4.1.2. A different test strip shall be used with each solution.

4.4.1.2 Test procedure.

a. Immerse reagent pad of test strip into a standard solution. Remove test strip immediately from solution and shake excess fluid from strip.
b. Wait 15 seconds.
c. Match the color obtained on the reagent pad to the color chart for determination of color reaction as indicated below:

Sample test strip immersed in standard solution		Color block designated on color chart specified in 3.4.
a	shall match	POOR
b	shall match	BORDERLINE
c	shall match	GOOD

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

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5.1.1 Unit of issue. Each test kit, consisting of one bottle containing twenty-five strips along with directions for use and a length of tubing, as specified, constitutes one unit of issue.

* 5.1.2 Level A.

* 5.1.2.1 Unit packaging. Each test kit shall be packaged in a box constructed in accordance with PPP-B-566 or PPP-B-676. Closure of box shall be in accordance with the appendix of the applicable box specification.

* 5.1.2.2 Intermediate packaging. Not more than twenty-four test kits shall be packaged in a box constructed in accordance with PPP-B-636, type CF, class domestic. Closure of box shall be as specified in the appendix of the box specification.

* 5.1.3 Level C. Each kit shall be packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

* 5.2.1 Level A. Not more than one hundred forty-four test kits shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

* 5.2.2 Level B. Not more than one hundred forty-four test kits shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

* 5.2.3 Level C. The packaged kits shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking.

5.3.1 Unit package. Each unit package shall be commercially identified and shall include the National Stock Number (NSN). The National Stock Number may be applied by a pressure-sensitive label. Marking shall include the following legends:

a. "CAREFULLY READ DIRECTIONS FOR USE ON INSERT BEFORE PERFORMING TEST".
b. "REPLACE CAP IMMEDIATELY".
c. "KEEP BOTTLE TIGHTLY CLOSED AND CONTENTS DRY".
d. "DO NOT TOUCH TEST END OF STRIP".
e. "DO NOT REMOVE DESICCANT FROM BOTTLE".
f. "25 TESTS".

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5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. Lot (control) number, contract number, and name of contractor shall be shown. Date of manufacture shall be shown in lieu of date packed. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.).

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Lot (control) number shall be shown. Date of manufacture shall be shown in lieu of date packed. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86°F.)".

6. NOTES

6.1 Intended use. Item covered in this specification is used in conjunction with NSN 6630-00-105-1418 Tester, Antifreeze and Battery,and Army Technical Bulletin TB 750-651.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1. and 5.2).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially availabe, but is intended to cover the types which are normally procured to meet military requirements.

6.4 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6630-00-169-1506	TEST KIT, Reserve, Alkalinity, Antifreeze

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EA
Navy - MS
Air Force - 99

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